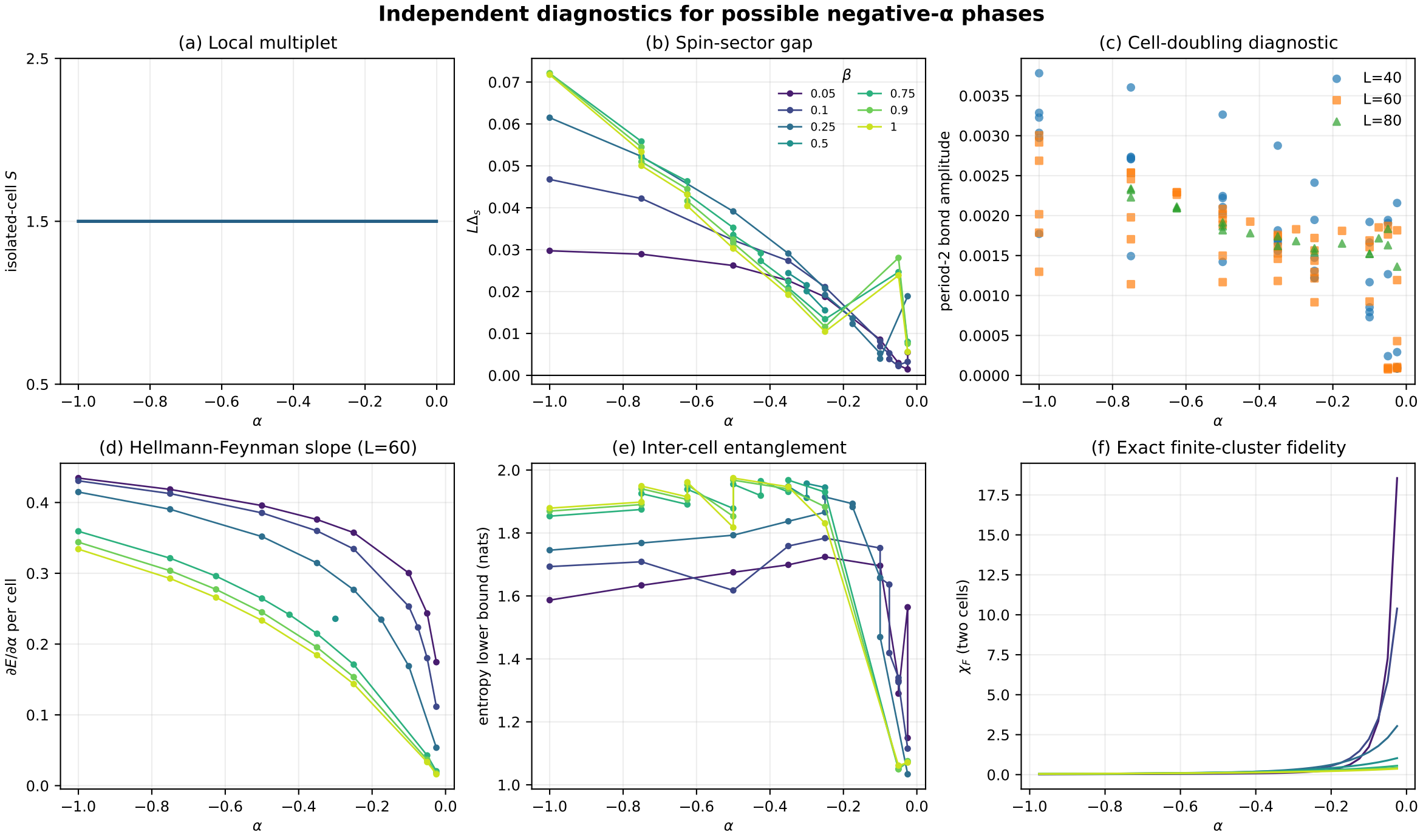
Independent diagnostics for possible negative-α phases
(a) Local multiplet
isolated-cell S
α
(b) Spin-sector gap
$L\Delta_S$
β
0.05
0.1
0.25
0.5
0.75
0.9
1
α
(c) Cell-doubling diagnostic
period-2 bond amplitude
L=40
L=60
L=80
α
(d) Hellmann-Feynman slope (L=60)
∂E/∂α per cell
α
(e) Inter-cell entanglement
entropy lower bound (nats)
α
(f) Exact finite-cluster fidelity
χ_F (two cells)
α